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Subject Name: Advanced Communication and Interpersonal Skills

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Semester: 2

Reading Comprehension (Intermediate Level)

Learning Objectives

- To **define** the main aspects of effective text comprehension.
- To **understand** the difference between the main idea and supporting details in academic and technical texts.
- To **use** reading strategies like **skimming and scanning** active reading strategies (Skimming and Scanning) to locate information efficiently.
- **Formulate** accurate, text-based answers to comprehension questions.
- **Identify** the central theme or argument of a given passage.
- **Develop** skills for summarizing short texts concisely and accurately.

What is Reading Comprehension?

The skill to read, process and comprehend the meaning of text is called **Reading Comprehension**.

Reading Comprehension employs context, linking ideas and logic to build meaning. The purpose is to comprehend the message and purpose of the author.

Why is it Critical for Engineers?

Engineers have to deal with various technical documents that require clear and accurate interpretation. Good comprehension skills enable Engineers to:

1. **Effectively read** Technical manuals and specifications.
2. Do a **focused** review of the research paper.

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3. Have a clear **understanding** of safety protocols and contract documents.

Note: A lack of comprehension skills may lead to misreading a text, ultimately resulting in errors and costly mistakes.

1. Key Concepts/Definitions

Concept	Definition	Type of Reading
Reading Comprehension	The cognitive process of constructing meaning from text.	Intensive Reading (for detail)
Skimming	Reading quickly to get the Main idea	Main idea or a general overview.
Scanning	Quickly looking for	Specific information or keywords.
Main Idea / Thesis	The central point the author wants to convey—the "umbrella thought".	N/A
Supporting Details	Facts, examples, reasons, and data that	Prove or elaborate the main idea.

2. Types of Texts

- **Expository:** Expository texts impart information, facts and concepts. The aim of expository texts is to deliver the information to a wider audience.
- **Descriptive:** Descriptive texts aim to deliver a clear image of an object, person or place through description.
- **Narrative:** Tells a story (e.g., a case study or a scientist's invention story).

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- **Persuasive:** Attempts to convince the reader (e.g., an editorial arguing for a renewable energy policy).

3. Finding the Main Idea

The **Main Idea** is what the author wants to say about the topic.

- **Where to Look:** The main idea (or thesis) is often found in the **Introduction** or the **Conclusion**. Within a paragraph, look at the **Topic Sentence** (first or last sentence).
- **Key Questions to Ask:**
 - What is the author's **primary purpose**?
 - What do **all the details** have in common?
 - What point is made **repeatedly**?

4. Understanding Supporting Details

Supporting Details clarify, prove, or elaborate the main idea. They answer the fundamental questions: Who? What? Where? When? Why? How?

Category (The 3 Es)	Function	Signal Words/Phrases
Examples	Illustrate and clarify the main point.	<i>For instance, such as, specifically</i>
Explanations	Provide context, cause, or background.	<i>Because, in other words, as a result</i>
Evidence	Offer concrete proof, data, or statistics (Most crucial in technical texts).	<i>Studies show, according to research, 75% of...</i>

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5. Strategy for Answering Questions Accurately

Accuracy is key in technical reading. Your answer must be supported by the text.

Step	Action	Focus
Analyze the Question	Underline key terms (e.g., <i>define, contrast, state the reason</i>) to identify the type of information needed (fact, cause, effect).	What specifically am I being asked?"
Locate the Answer	Scan the passage for the keywords. Read the sentences containing the keywords carefully.	Where does the text support my answer?
Formulate the Response	Answer only what is asked. Use complete sentences. Paraphrase the text unless a direct quote is needed. Do not add external knowledge.	Is my answer fully supported by the passage?

6. Types of Comprehension Questions:

- **Literal:** The answers are directly stated in the text.
- **Inferential:** Inferential questions require "read between the lines" to determine the author's implied meaning or purpose.
- **Critical:** Requires evaluation or opinion, which must still be backed by evidence from the text.

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7. The S.C.A.N. Method: Your Quick Check for Text Comprehension

Mnemonic	Full Meaning	Detailed Action
S	SKIM & SPOT	Quickly read the title, headings, and key sentences to spot the main idea
C	CLUE IN	Clue in on the Question by analyzing the action verbs and keywords to know what to search for.
A	ANALYZE & ANCHOR	Scan the text for keywords. When found, anchor your reading to the surrounding sentences to identify the exact details.
N	NARROW & NAIL	Narrow the answer to <i>only</i> text information. Nail it by writing a complete, accurate answer

8. Real-life Applications/Case Examples

- When reviewing a tender document, an engineer should skim the Executive Summary to understand the main selling points and scan the Bill of Quantities to find the cost of a specific component. This helps in preparing a counteroffer.
- Debugging Code (Intensive/Literal Reading): Reading a coding tutorial line by line or intensively reading a machine's fault codes requires Literal comprehension—understanding exactly what each line or code means, with zero inference.
- Patent Analysis (Expository/Descriptive Text): An engineer needs to analyze the description of a new invention to determine if it infringes on existing patents, requiring a clear distinction between the inventor's claim (Main Idea) and the specific process steps (Supporting Details).

9. Summary/Key Takeaways

- Reading is an active skill, not a passive one. It requires strategic effort (Skimming and Scanning).

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- Always distinguish the Main Idea (the point) from Supporting Details (the proof).
- Answer questions by strictly referencing the text – No Text, No Answer.
- Comprehension skills (finding theme, summarizing, vocabulary) are crucial for efficient knowledge absorption in your engineering

10. References:

Anderson P. V. *Technical Communication*. Wadsworth, 1998.

Alley M. *The Craft of Scientific Writing*. Springer, 2018.

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